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1302 Project Report

The COVID-19 pandemic has had a significant impact on many parts of the world, including Los Angeles County, California. To gain a better understanding of the virus's impact in the area, this report presents an analysis of a dataset containing information about the number of confirmed cases, deaths, and hospitalizations due to the virus. The data was collected daily from various sources and is presented in a CSV file. The code reads and processes the data, including converting the date column to datetime format, sorting the data by date, and dropping unnecessary columns.

After processing the data, the code calculates several statistical measures such as mean, median, variance, and standard deviation for the number of cases and deaths at both the county and state levels. This information provides a better understanding of the pandemic's trend in the area and helps to identify potential areas of concern.

To visualize the data, the code creates several visualizations. A line plot depicts the running number of cases in Los Angeles County and California over time, showing the pandemic's overall trend in the area. The plot highlights the difference in the number of cases between Los Angeles County and the state of California.

A pie chart is also included, showing the proportion of new cases vs. new deaths in LA County. This chart is useful in understanding the severity of the pandemic, with new cases comprising a larger proportion of the total than new deaths.

Furthermore, a bar plot is included comparing the number of new cases vs. new deaths in LA over time. This plot allows us to see trends in new cases and deaths over time and identify any emerging patterns. The plot shows that while the number of new cases has decreased, the number of new deaths has remained relatively stable.

Lastly, a scatter plot is included, comparing the number of new deaths in Los Angeles with the number of new deaths in California over time. This plot allows us to see if there is any correlation between the two variables and whether there is a difference in the pandemic's trend between Los Angeles and California as a whole. The plot shows that the number of new deaths in Los Angeles is generally lower than the number of new deaths in California, and there is a slight correlation between the two variables.

To sum up, this report analyzed a dataset related to the COVID-19 pandemic in Los Angeles County, California. By processing and analyzing the data using statistical measures and visualizations, the report sheds light on the pandemic's overall trend in the area and identifies potential areas of concern. The visualizations help to better understand the data and identify emerging patterns. In conclusion, this analysis provides valuable insights into the impact of the COVID-19 pandemic in Los Angeles County and the state of California as a whole.